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Enoch Hyunwook Kang

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Research Interests

Computer Science methods for social science: AI agents, AI alignment, Machine learning theory, Reinforcement learning theory, Game theory, Adaptive experimentation theory, Graph neural networks

Econometric methods for social science: Dynamic Structural Models, Double/Debiased Machine Learning

Education

University of Washington, Seattle
Ph.D. Candidate, Marketing, Expected 2027; M.S., Marketing, 2025

Texas A&M University, College Station
Ph.D., Computer Engineering, 2023
Dissertation: 'Efficiency and robustness in online learning for personalized experiments and recommender systems'

Korea Advanced Institute of Science and Technology (KAIST), Daejeon
B.S., Mathematics, 2017
B.E., Industrial & Systems Engineering, 2017

Publications

1. Kang, E. H., and Kumar, P. R., (2024) "Is O(log N) Practical? Near-Equivalence Between Delay Robustness and Bounded Regret in Bandits and RL," ***Advances in Neural Information Processing Systems (NeurIPS)***.

2. Kang, E. H., and Kumar, P. R., (2023) "Bounded (O(1)) Regret Recommendation Learning via Synthetic Controls Oracle," ***Allerton Conference on Communication, Control, and Computing***.

3. Kang, E. H., Kwon, T., Morrison, J. R., and Park, J., (2022) "Learning NP-Hard Multi-Agent Assignment Planning using GNN: Inference on a Random Graph and Provable Auction-Fitted Q-learning," ***Advances in Neural Information Processing Systems (NeurIPS)***.

Working Papers

4. Kang, E. H., Jain, L., and Yoganarasimhan, H., (2025) "Empirical Risk Minimization for Inverse RL and Dynamic Discrete Choice Models," Invited for Minor Revision at ***Operations Research***.

    Main Track, Economics and Computation (EC), 2025.

    Invited tutorial talk (with John Rust), Econometric Society Summer School in Dynamic Structural Econometrics, 2025. Title: "Structural Estimation and Reinforcement Learning"

5. Kang, E. H., and Yoganarasimhan, H., (2025) "TextBO: Bayesian Optimization in Language Space for Eval-Efficient Self-Improving AI," *Working Paper*.

6. Kang, E. H., and Jang, K., (2025) "Stability and Generalization for Bellman Residuals," *Working Paper*.

7. Kang, E. H., (2025) "LLM Personas as a Substitute for Field Experiments in Method Benchmarking," *Working paper*.

8. Park, B., Park, M., Kang, E. H.*, and Jang, K.*, (2025)"Globally Convergent Offline Reinforcement Learning with Smoothed Bellman Residual Minimization," *Working paper.*

9. Kang, E. H., (2026) "Reasonably Reasoning AI Agents Avoid Game-Theoretic Failures in Zero-Shot, Provably," *Working Paper*.

10. Kang, E. H., (2026) "Demystifying the unreasonable effectiveness of online alignment methods," *Working Paper*.

11. Kang, E. H., (2026) "Personalized Alignment Revisited: the Necessity and Sufficiency of User Diversity in Choice Modeling" *Working paper*.

12. "Multi-turn alignment revisited", *Working Manuscript*.

13. "Test and Self-Improve for Markov Sufficiency: An AI-driven Dynamic Measurement Framework for Long-Term Outcomes" (Project with Adobe), *Working Manuscript*, .

14. "Causally Debiasing Unstructured Data using Representations In Language Models," *Working Manuscript*.

Book chapters & Lecture notes

1. Kang, E. H., (2026) "A Lecture Note on Offline RL and IRL, Part I: Foundations of Offline Reinforcement Learning "

2. Kang, E. H., (2026) "A Lecture Note on Offline RL and IRL, Part II: Foundations of Inverse Reinforcement Learning and Dynamic Discrete Choice Models "

Service

Reviewer

Marketing Science
ACM Economics and Computation (2026)
ICLR (2026, 2025, 2024, 2023)
ICML (2026, 2025, 2024, 2023)
NeurIPS (2026, 2025, 2024, 2023, 2022)
AISTATS (2023)

Invited Talks

Empirical Risk Minimization for Inverse RL and Dynamic Discrete Choice Models.

Econometric Society Summer School in Dynamic Structural Econometrics, 2025
(Tutorial Talk (with John Rust), "Structural Estimation and Reinforcement Learning")
UBC Econometric Lunch Seminar, 2025

Conference Presentations

Reasonably Reasoning AI Agents Avoid Game-Theoretic Failures in Zero-Shot, Provably.

ICLR MALGAI Workshop, 2026 (Poster)

Stability and Generalization for Bellman Residuals.

ICML Decision-Making from Offline Datasets to Online Adaptation Workshop, 2026

Globally Convergent Offline Reinforcement Learning with Smoothed Bellman Residual Minimization.

ICML Decision-Making from Offline Datasets to Online Adaptation Workshop, 2026

LLM Personas as a Substitute for Field Experiments in Method Benchmarking.

ICML CTB (Combining Theory and Benchmarks) Workshop, 2026
Advances with Field Experiments (AFE) Conference, 2026

TextBO: Bayesian Optimization in Language Space for Eval-Efficient Self-Improving AI.

Stanford AI & Marketing: New Methods and New Risks Conference, 2025
Columbia AI/ML Conference, 2025
Frank M. Bass UTD-FORMS Conference, 2026
Symposium on Artificial Intelligence in Marketing, 2026
ICLR AI with Recursive Self-Improvement Workshop, 2026 (Poster)

Empirical Risk Minimization for Inverse RL and Dynamic Discrete Choice Models.

ACM Economics and Computation (EC), 2025
The World Congress of the Econometric Society (ESWC), 2025

Is O(log N) Practical? Near-Equivalence Between Delay Robustness and Bounded Regret in Bandits and RL
NeurIPS, 2024
NeurIPS ReALML Workshop, 2023 (Poster)

Recommender System as an Exploration Coordinator: A Bounded O(1) Regret Algorithm for Large Platforms
Allerton Conference, 2023 (Oral)
ICML ReALML Workshop, 2022 (Poster)
RecSys Causality & Sequential Decision-Making Workshop, 2022 (Long Oral)

Learning NP-Hard Multi-Agent Assignment Planning using GNN: Inference on a Random Graph and Provable Auction-Fitted Q-learning.
NeurIPS, 2022
ICML GRL+ Workshop, 2022 (Oral)

Honors and selections

Dean's achievement award, Foster School of Business, 2026

Selected Proposal, MSI – Adobe Initiative
"Test and Self-Improve for Markov Sufficiency: An AI-driven Dynamic Measurement Framework for Long-Term Outcomes", 2026

Selected Scholar (1 of 30 faculty and PhD candidates), AI and Economics (AIE) Summer Institute, Center for Applied Artificial Intelligence, Chicago Booth, 2026

Gold Reviewer (Top 25%), International Conference on Machine Learning (ICML), 2026

Best Paper Runner-Up, ICML GRL+ Workshop, 2022

Immigration Status

U.S. Permanent Residency (Green Card) Pending: Currently under I-130 (Petition for Alien Relative) processing.